

LLMs, Prompt Engineering & AI Development Platforms

Prompt Engineering

➤ What is Prompting

It is a short pieces of text that are used to provide context and guidance/ instructions to ML models to get desired output.

➤ How LLMs Understand Prompts

- First it break sentence into small units. It is called Tokens
- Tokens transform into vectors(numbers) similar meaning have close vectors it is called “semantic space” it is called embeddings
- Transformer Attention uses to see relationship b/w tokens in entire prompt .it help to understand the sentence
- Prediction , it will predict the next word based on training
- Then it will generate output
- Modern LLMs can also understand context, instructions, and multimodal inputs

➤ Role Prompting

Assigning a person or role to a Large Language Model (LLM) to shape its responses , before asking a question.

- The purpose it will help you to give better answer , responses will match it will give better text
- For good prompt includes ROLE , TASK, CONTEX , OUTPUTFORMAT
- EX: “You are a research assistant. Summarize this report clearly for a high-school audience in under 200 words. Return as three bullet points and one summary sentence.”

➤ Context Prompting

Context prompting means giving an AI **extra background information** along with your main question or request, so the answer is more accurate, relevant, and useful.

- It will understand the situation better and give better answer so LLMs maintain consistency
- It matters clarity , relevant , quality

EX: **without context**: "Write me a poem."

With context : "Write me a short, funny poem for my best friend's birthday party."

➤ Structured Prompting

Structured prompting **means organizing your request in a clear step-by-step way** so the AI understands exactly what you want.

- It helps to give clarity , focus, better results
- If we want ask anything AI we try to include **Who** it's for (audience), **What** you want (task), **Why** you need it (purpose) **How** it should look (tone, style, length, etc.) it will provide better results

EX: **Unstructured Prompt**: "Write a story."

Structured Prompt: "Write a short, funny poem (tone) for my best friend (audience) to read at their birthday party (purpose). Keep it under 8 lines (length)."

Advanced Prompting

➤ Chain-of-Thought Prompting

Chain-of-thought prompting is a way of asking an AI to **show its thinking step by step** before giving the final answer.

- It help to show its reasoning process and reduces mistakes in complex questions.
- This method is useful for **math problems, logical reasoning, coding, and decision-making tasks**. Modern LLMs improve accuracy and reasoning performance

Ex: **Unclear Prompt:** “What is 12×15 ?” The AI might just say: “180.”

Chain-of-Thought Prompt: “Solve 12×15 step by step.”

$$12 \times 10 = 120$$

$$120 \div 5 = 60$$

$$120 + 60 = 180 \text{ Final Answer: } \mathbf{180}$$

➤ Few-Shot Prompting

Few-shot prompting means giving the AI **a few examples** of what you want before asking it to do the task.

Strategies

- Use clear, relevant examples aligned with the task.
- Include diverse examples to improve generalization.
- Use few high-quality examples to avoid noise.
- Test and refine prompts regularly.
- Avoid overfitting by testing on new inputs.

Applications

- Content creation
- Video action recognition
- Translation
- Chatbots

Advantages

- It requires minimal training data, enables the fast task learning , done multiple task easily , reduce the data and processing cost, Useful where large datasets are unavailable.

Limitations

- In some cases it will give incorrect answers
- Less effective for complex tasks.
- Can introduce bias from limited examples.
- Risk of overfitting to given samples.

➤ Multimodal Prompting

Multimodal prompting means using **different types of input together** (like text, images, audio, or video) to ask the AI for help. Instead of only words, you can mix other modes (like showing a picture + asking a question).

- **It works** , user give multiple inputs and AI will connect it together then it provide answer
- **Need** of this is when text only is not enough , pictures and audio will help for extra context it provide accurate answer
- We use it for **Education, Healthcare ,Shopping, creative work**
- When text is only not enough to get clear answer , then we this model
- Ex: Here's a photo of a plant. Tell me if it looks healthy.

➤ Hallucinations

- **Hallucination** occurs when an AI model generates information that sounds correct but is actually **wrong, false, or made up**.
- The AI may create fake facts, references, names, dates, or answers when it is uncertain.
- **It happens** because of Incomplete context , Overgeneralization (rely on training data) , Ambiguity (unclear), Bias in training data, Pressure to be helpful

Types of Hallucinations:

- **Factual hallucinations:** Incorrect dates, names, statistics, or references.
- **Logical hallucinations:** Flawed reasoning or contradictions within the response.
- **Contextual hallucinations:** Irrelevant or tangential information that doesn't fit the user's request.
- **Semantic hallucinations:** Misuse of terminology or invented jargon.

REDUCE THE HALLUCINATIONS : Use techniques like **Chain-of-Thought**, **Few-Shot Prompting**, and **Retrieval-Augmented Generation (RAG)**.

EX: **Weak Prompt:** "Tell me about the history of quantum computing." →
Risk: The model may invent milestones or misattribute discoveries.

Strong Prompt: "Provide a timeline of quantum computing milestones from verified academic sources. Include citations and avoid speculative claims." →
Result: More accurate, grounded, and less prone to hallucination.

Applied Generative AI & Intelligent Systems

Llama

- Llama (Large Language Model Meta AI) it is a family of AI models created by Meta (the company behind Facebook).
- It is a **Generative AI system** that can understand and create text, images, and even analyse documents.
- The newest version, **Llama 4 (2026)**, is very powerful and open-source, meaning anyone can use it freely.
- We use for Learning and research , Coding help , Document analysis , Creative work
- Through **apps and platforms** like Azure AI Foundry or Grok Cloud. By giving it **prompts** (instructions in plain language).

ADVANTAGES: Open Source(free) , **Efficient** (uses fewer resources). Multimodal , **Long memory** (reads huge documents).

DISADVANTAGES: Some versions still training. Lower accuracy in complex math/coding. Needs strong computers for big models.

Limitations : Not always 100% correct (hallucinations). Needs clear prompts. Limited real-time updates.

Updated Info:

- **Llama 4 Scout:** runs on small computers, 10M token memory.
- **Llama 4 Maverick:** strong multimodal reasoning, beats GPT-4o in some tests.
- **Llama 4 Behemoth:** still training, may rival GPT-4.5 and Gemini Ultra.

RAG (Retrieval-Augmented Generation)

- It is a way to make AI smarter by letting it **search for information first** and then **generate answers**.
- We use this because **reduce mistakes, up-to-date answers, AI more trustworthy and useful** in real-world tasks.

HOW WE USE IT ?

- Connect AI to the database or document store . “Explain Newton’s laws from my notes”, RAG retrieves the notes first, then explains.
- Tools used to build RAG system are Long Chain, LlamaIndex, Haystack

Advantages : **Accurate answers** with citations. **Up-to-date knowledge** from external sources. **Works across subjects** (science, law, medicine). **Reduces hallucinations** compared to plain AI.

Disadvantages: Needs **good databases**; poor data = poor answers. **Complex setup** compared to simple AI. **Slower responses** because it searches first.

Limitations: still it is not perfect , it want clear queries , costly for big system

Updated info :

Hybrid RAG: combines vector search, keyword search, and graph search for better accuracy.

Agentic RAG: AI acts like an agent, breaking complex questions into smaller steps.

Multimodal RAG: can retrieve not just text but also images, tables, and structured data.

- **Companies like DataRobot AI, Algoworks, and BeyondKey** are building advanced RAG solutions for businesses.

YOLO

YOLO stands for **You Only Look Once**. It is an **object detection AI model** used in Computer Vision. YOLO can identify and locate objects in an image or video in a single step.

- It draws **bounding boxes** around objects and labels them (person, car, dog, etc.). Modern versions like **YOLOv11 and YOLOv12** are faster and more accurate than earlier versions.
- WE USE this for **find and recognize objects quickly. real-time tasks** like self-driving cars or CCTV monitoring. **It saves time**

Where We Use It: Self-driving cars, Security cameras, Healthcare (detect tumor), Robotics

- How We Use It ? Train YOLO with **datasets** (lots of labeled images). Run it on a computer with a camera or video feed. It draws **bounding boxes** around detected objects and labels them.
- Image → Detect Objects → Draw Boxes → Show Results
- **Advantages :** Fast , Accurate , Efficient , Versatile
- **Disadvantages:** Struggles with **small objects** , Needs **large** datasets for training. **Complex setup** for beginners.
- **Limitations :** Cannot always detect objects in poor lighting or blurry images. Accuracy depends on training data quality. Requires powerful hardware for advanced versions.
- Updated Information : **YOLOv11 and YOLOv12** these are updated models it is better speed and accuracy , it will support drones , mobile phones , IOT cameras , it will work with **AI agents, robotics, and multimodal AI applications**.
- Used in smart surveillance, autonomous vehicles, healthcare imaging, and industrial automation.

EX: CCTV monitoring

➤ Whisper

- **Whisper** is an AI speech recognition model developed by **OpenAI**.
- It converts **spoken audio into text** (Speech-to-Text).
- It can also help with translation from one language to another.
- Whisper is widely used in AI assistants, transcription tools, and voice applications. It is combining with LLMs to create voice based AI assistant
- Helps people who want to **transcribe lectures, meetings, or interviews**. Useful for **language learning people**

How We Use It

- Install Whisper or use it through apps.
- Speak into a microphone → Whisper listens → it produces text.
- Can also translate speech from one language to another.

Advantages: **Accurate** across many languages. **Free and open-source** (anyone can use it). **Supports translation** (speech-to-text in another language). **Works offline** if installed on a computer.

Disadvantages : Needs **powerful hardware** for fast performance
. **Large files** (takes up storage). May struggle with **noisy environments**.

Limitations : Not perfect with **rare languages or accents**. Cannot always handle **background noise** well. **Does not understand meaning** — only converts speech to text.

Updated info : Whisper v3 (2026): faster, smaller, and more accurate than older versions. **Edge Whisper:** runs on mobile phones and IoT devices. **Multimodal Whisper:** combines speech recognition with video analysis (e.g., captions + scene detection). Used in **education apps, smart assistants, and healthcare transcription systems**.

AI Agents (LangChain / CrewAI)

- **AI Agents are smart programs that can act like helpers.** They don't just answer questions — they can **plan, search, and take actions** step by step.
- **LangChain** and **CrewAI** are popular frameworks to build these agents.
- **Used in** smart education apps, enterprise automation, and AI research labs.

LangChain: IT is a framework used to **connect LLMs with tools, databases, APIs, and documents**. It helps **create AI applications** that can perform multiple tasks.

- LLM + Tools + Data = LangChain Application

CrewAI : CrewAI **allows multiple AI agents to work together** as a team. Each agent has a different role and responsibility.

How We Use Them? Build agents using **LangChain** or **CrewAI**. Give them **roles** (like “teacher,” “researcher,” “planner”). They use **tools** (search engines, calculators, databases) to complete tasks.

Advantages : Multi- step reasoning , **Tool use** (search, code, databases). Teamwork , **Flexible** (can be customized for different jobs).

Disadvantages: Complex setup compared to simple AI. Slower because they plan and search. Needs good data — wrong data leads to wrong answer.

Limitations: Still can **hallucinate** if not grounded. **Expensive** for large-scale use (computing power). **Hard to control** if agents act unpredictably.

Updated info:

LangChain 2026: supports multimodal agents (text + images + audio).

CrewAI 2026: improved teamwork features, agents can debate and choose best answers.

Agentic RAG: combines retrieval with agents for more accurate results.

AI Development Platforms

A place where programmers build, train, and test artificial intelligence applications.

Ex: Google AI, Microsoft Azure AI, GitHub

GitHub:

- **GitHub** is a cloud-based platform used to **store, manage, and share code**. It is built on **Git**, a version control system.
- Developers use GitHub to collaborate on software and AI projects.
- It acts like a **Google Drive for code**, but with extra features for tracking changes. This help to keep their code safely
- Today, GitHub is one of the most important platforms for **AI and software development**.
- Create Repo → Upload Code → Commit Changes → Collaborate → Deploy

Why we USEs: Collaboration , Version Control(avoid mistakes) , AI Assistance , Security (fixes code)

Where we use: Software projects , AI projects , Education , Companies(Used by startups and big enterprises like Duolingo and Accenture)

Advantages: Teamwork , AI-powered coding , Free access , Security tools

Disadvantages: Internet required , Complex for beginners , Private projects cost , AI limitations(sometimes chatbots will give wrong answer)

Hugging Face : **Hugging Face** is a popular AI platform where developers can find, use, train, and share AI models.

- It is often called the "**GitHub of AI**" because it hosts thousands of AI models and datasets.
- It provides models for **text, image, audio, video, and multimodal AI**. Many modern AI applications use models available on Hugging Face. It use libraries like **Transformers** in Python for coding.

We Use Hugging Face : Free access , Collaboration , Pre-trained models , Multi-modal support

Advantages: Open-source , **Large library** (Millions of models available), **Easy integration** (Works with Python and other languages) , **Enterprise support**(Paid plans for companies with security features)

Disadvantages: Cannot use offline , Running large models needs powerful GPUs, which can be expensive , Complexity , Public datasets may not suit sensitive projects.

Ollama

- It is an application built on **llama.cpp** that allows you to run **Large Language Models (LLMs)** locally / in your computer.
- it helps users use AI models such as **Llama, Mistral, Gemma, DeepSeek, and Qwen** locally.
- It connects with Hugging Face's **Model Hub**, where thousands of models are stored.
- The use of this is you no need to send your data to cloud
- To keep data private and secure. without internet dependency.
- To reduce cloud API costs. Offline mode
- Normally: Question → Internet → Cloud AI → Answer

- With Ollama: Question → Your Computer → Local AI Model → Answer
- How do you use ?
Install Ollama → Download Model → Run Model → Ask Questions
- Model are Llama , Mistral , Gemma , DeepSeek

Advantages: Privacy-first , Offline operation , Cost efficiency , Flexibility(hugging face model)

Disadvantages: Large models require strong CPUs/GPUs. Beginners may find installation tricky. Local models may run slower than cloud AI. Very large models may not fit on personal devices.

LangChain

- **LangChain** is an open-source framework used to build AI applications and AI Agents.
- It helps connect **Large Language Models (LLMs)** with tools, databases, documents, APIs, and websites.
- Instead of just answering questions, LangChain helps AI perform tasks and workflows.
- It is widely used for **chatbots, RAG systems, AI assistants, and AI agents**. It includes LangGraph (for orchestration) and LangSmith (for debugging and monitoring agents).
- **Use this langchain** build advanced AI applications , connect AI models with external data , develop RAG system , create AI agent that can perform task
- User Question → Get Data → Send to LLM → Generate Answer it works like this
- **Advantages** include flexibility and scalability, while limitations include complexity and hardware needs.
- **Disadvantages :** Complexity , Hardware needs, Debugging challenges , Performance limits

Vibe Coding

Vibe Coding is a way of creating software by describing what you want in simple natural language, and AI generates the code. It is also called AI-Assisted Development.

- AI Coding Tools : GitHub Copilot , cursor AI , OpenAI ChatGPT , Google Gemini , claude AI
- It helps developers create applications faster and improves productivity.
- AI assists with coding, but humans must still review, test, and improve the final software.

AI Assisted Development

AI-Assisted Development means using AI tools to help developers write, understand, test, and improve code faster.

- We use this for Speed , Learning , Error reduction , Creativity
- How we use? Install an AI coding assistant , Open your coding editor , Start writing or describing what you want , Accept, edit, or reject AI's suggestions , Test and improve with AI debugging tools
- **Advantages:** Saves time , Explains code in simple terms , Reduces mistakes , Lets you focus on ideas
- **Disadvantages:** people will depend on AI , sometimes it will give wrong information , advance features may cost money
- **Ex:** "Create a Python program that asks for the user's name and displays a welcome message."

AI Coding Tools

- AI Coding Tools are applications that use AI to make **coding easier**
- Examples: **GitHub Copilot, Hugging Face Transformers, LangChain, Tabnine.**
- They can autocomplete code, explain errors, and even generate entire functions.
- We use this because **Speed , Learning , Error reduction , Creativity .**

- How to use ? Install the tool , write what you want , AI will suggest code , accept the code if error is there ask to debug with AI help
- Advantages and Disadvantages same as AI Assisted developer

Cursor

- Cursor is a coding editor (like VS Code) but with built-in AI tools.
- It uses models like **GPT-4, Claude, and Llama** to suggest code, explain errors, and even generate entire projects. guides **AI-assisted suggestions**
- **We use it for** Shows where code will appear , it help to suggest the code , Lets you insert, delete, or replace code easily.
- In coding editors , data bases , terminals (shows commands will be types) we use it here
- **Advantages :** Clear guidance , AI-friendly , Easy editing
- **Disadvantages:** Confusion , Accessibility issues , Database performance , Dependence